INCH-POUND

A-A-50796A
12 December 2011
SUPERSEDING
A-A-50796
5 May 1982

COMMERCIAL ITEM DESCRIPTION

SIREN, ELECTRONIC

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers electronic sirens for use on law enforcement and emergency response vehicles.

2. CLASSIFICATION. The siren shall be classified by the following types, classes, and styles:

2.1 Type. The siren shall be one of the following types (see 7.3(b)):

Type I - Complete siren, including an amplifier with attached microphone, amplifier mounting bracket and hardware, microphone mounting clip, interconnecting wire harness, speaker(s), and speaker mounting bracket(s) and hardware
Type II - Amplifier with attached microphone, amplifier mounting bracket and hardware, microphone mounting clip, and interconnecting wire harness

2.2 Class. The siren shall be one of the following classes (see 7.3(c)):

Class A - Operates on a nominal 12 volt power supply
Class B - Operates on a nominal 24 volt power supply

2.3 Style. The siren shall be one of the following styles (see 7.3(d)):

Style 1 - Single-speaker system
Style 2 - Two-speaker system

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or DLA Aviation VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.daps.dla.mil/.

FSC 6350

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3. SALIENT CHARACTERISTICS.

3.1 Design and construction. The siren shall be of compact design and sufficiently rugged to withstand shock and stress incidental to use on an emergency response vehicle.

3.1.1 Overall Dimensions. The maximum overall dimensions of the amplifier, including the mounting bracket, shall be not more than eight inches wide, five inches high, and nine inches deep.

3.1.2 Materials. The amplifier housing shall be made of 18 gauge steel or heavier to provide strength and electromagnetic shielding. Metals used in the manufacture of the siren equipment shall be either corrosion resistant or treated to prevent corrosion. The use of dissimilar metals, especially brass, copper, or steel in intimate contact with aluminum or aluminum alloy shall be avoided. Protective coatings which will crack, chip, peel or scale with age shall be avoided. When materials are used that are nutrients for fungi and they are not hermetically sealed, they shall be treated with a fungicidal agent. The siren shall be wired with a type of wire which conforms to MIL-DTL-16878/1, "Wire, Electrical, Polyvinyl Chloride (PVC) Insulated, 105 °C, 600 Volts (Not for Navy Shipboard Use)".

3.1.3 Components.

3.1.3.1 Amplifier bracket. The amplifier assembly with mounting bracket shall be designed for mounting above or under the dashboard. The amplifier mounting bracket shall allow the amplifier to be readily removed for repair purposes.

3.1.3.2 Power indicator. The amplifier's front panel illumination and/or power-on indication shall be by a non-glare pilot light.

3.1.3.3 Power switch. The amplifier's power shall be controlled by an on-off switch located on the volume control, or by a separate on-off switch.

3.1.3.4 Microphone. The microphone shall be a magnetic or noise cancelling carbon type.

3.1.3.4.1 Microphone cord. The microphone cord shall be the coil type and shall have an extended length of 8 feet ±1 foot.

3.1.3.5 Speaker system elements. Speakers and mounting brackets, when mounted on the exterior of the vehicle, shall be able to withstand a relative wind velocity of 100 miles per hour.

3.1.3.5.1 Speaker brackets. The speaker mounting bracket shall be designed to position the speaker on the vehicle centerline (if one speaker). The speaker bracket shall be designed in such a manner as to prevent easy removal by vandals.

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3.1.3.5.2 Speaker wires. Interconnecting speaker wire shall be, at minimum, 18 AWG stranded wire conforming to the requirements of 3.1.2, and a minimum of 15 feet long, unless otherwise specified (see 7.3(e)).

3.1.4 Circuit design. The electronic siren shall be constructed to use a floating ground return and shall not use the chassis for ground return. The siren shall not be required to operate on reversed polarity power connections, but shall not be damaged by such connections.

3.1.4.1 Power isolation. The amplifier's output shall be isolated from the primary input power by a transformer to prevent shorting the battery or damaging the internal components of the electronic siren in case the output becomes grounded.

3.1.4.2 Circuit protection. Internal circuit protection shall be provided by a fast- blow fuse (15 amps) or circuit breaker.

3.1.4.3 Operating current requirements.

a. Class A siren.
    (1) Maximum standby current shall be 0.5 amp.
    (2) With one speaker (driver), the maximum current shall be 6 amps.
    (3) With two speakers (drivers), the maximum current shall be 12 amps.
b. Class B siren.
    (1) Maximum standby current shall be 0.6 amp.
    (2) With one speaker (driver), the maximum current shall be 4 amps.
    (3) With two speakers (drivers), the maximum current shall be 6 amps.

3.1.4.4 Public address (PA) system. The minimum input required for operation of the PA system shall be 0.55 volt root-mean-square (rms) into an average load of 8 ohms.

3.1.5 Electromagnetic compatibility. Siren equipment provided to the government under this CID shall be compatible with radio communications equipment used on first-responder vehicles.

3.1.6 Environmental requirements. The siren shall be operable in the temperature range from -40 °C (-40 °F) to 70 °C (158 °F) and shall be designed to withstand a relative humidity up to 100 percent. The siren shall be designed to withstand vibration for a period of at least 1 hour at 12 Hertz (Hz) ±1 Hz, 0.3 centimeter (0.125 inch) peak-to-peak displacement. All speakers shall be weatherproof.

3.1.7 Modes of operation. The following modes of operation shall be selectable by controls on the front panel of the amplifier assembly. These modes shall be clearly marked as indicated:

a. Public address (PA) mode. This mode connects the microphone to the amplifier for use as a public address system. This mode shall be marked as "PA".
b. Radio mode. This mode amplifies calls received on a vehicle two-way radio through the PA system. This mode shall be marked as "RADIO".

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c. Manual mode. This mode enables the operator to manually control the siren pitch. This mode shall be marked as "MANUAL".
d. Wail mode. In this mode, the siren produces continuous automatic cycling of increasing and decreasing sound level as specified in 3.2(d). This mode shall be marked as "WAIL".
e. Yelp mode. In this mode, the siren produces a continuous automatic cycling of increasing and decreasing sound level as specified in 3.2(e). This mode shall be marked as "YELP".

3.2 Performance requirements. Siren sound pressure level requirements shall be verified in accordance with the methods specified in ANSI S1.13, "Measurement of Sound Pressure Levels in Air".

a. Sound levels. Measurements shall be made with 11-ohm, 75-watt speakers/drivers.
b. PA sound output. All measurements shall be made using a 1 volt rms test tone at 1000 Hz (see 7.4). With one speaker, the minimum PA sound output shall be 112 decibels (dB) at 10 feet. With two speakers, the minimum PA sound output shall be 116 dB at 10 feet.
c. Siren sound output. With one speaker, the minimum siren sound output shall be 114 dB at 10 feet. With two speakers, the minimum siren sound output shall be 118 dB at 10 feet. The siren output shall be a symmetrical square wave. The lowest fundamental frequency shall be between 250 Hz and 750 Hz. The frequency shall rise at least 900 Hz.
d. Wail. The siren shall produce a continuous automatic cycling of increasing and decreasing sound levels at a rate of 7-15 cycles per minute.
e. Yelp. The siren shall produce a continuous automatic cycling of increasing and decreasing sound levels at a rate of 100-225 cycles per minute.
f. Audio frequency response. The audio frequency response shall be flat within 3 dB from 250 to 4,000 Hz, at rated power. The zero dB reference shall be measured at 1,000 Hz.

3.3 Identification markings. The amplifier boxes shall be permanently marked with the manufacturer's identification and the model designation in accordance with MIL-STD-130, "Identification Marking of U.S. Military Property". Drivers shall be marked to denote manufacturer, model, and power rating in accordance with MIL-STD-130.

3.4 Workmanship. The finished siren and individual accessories shall be free from burrs, slivers, sharp edges, dents, and fractures. All cords, cables and interconnecting wire harnesses shall be free from kinks, nicks, cuts, burned or melted insulation, and excessive solder wicking. Bolts, nuts and screws shall not be missing, loose, fractured, or stripped.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

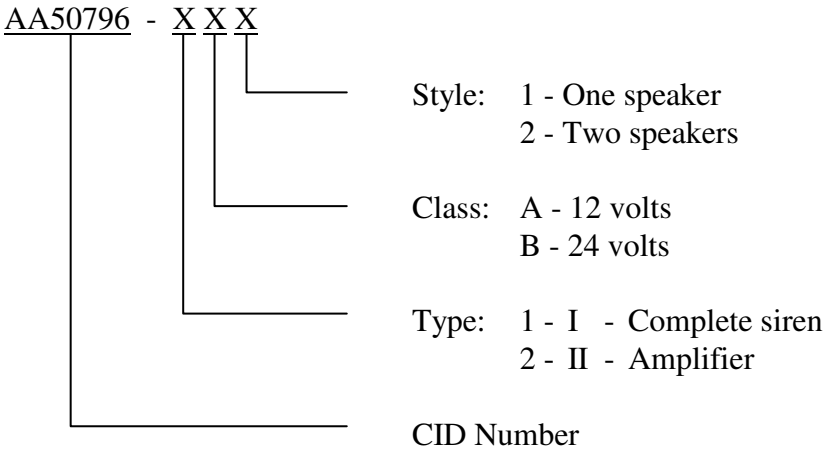
5.2 Market acceptability. The products offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING.

6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the acquisition order (see 7.3(f)).

7. NOTES.

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



Example of reference part number: AA50796 - 1 A 1

7.2 Sources of documents.

7.2.1 FAR. The FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies may be obtained from https://www.acquisition.gov/far/.

7.2.2 Military specifications and standards. Copies of military specifications and standards may be obtained from Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Electronic copies may be obtained from https://assist.daps.dla.mil/.

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7.2.3 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 25 West 43rd Street, 4th Floor, New York, NY 10036. Electronic copies may be obtained from http://web.ansi.org/.

7.3 Ordering data. The acquisition order should specify the following information:

a. CID document number, revision, and CID PIN.
b. Type (see 2.1).
c. Class (see 2.2).
d. Style (see 2.3).
e. Alternate speaker wire requirements (see 3.1.3.5.2).
f. Preservation, packing, and marking (see 6.1).

7.4 Hearing safety. Test personnel are cautioned to wear hearing protection when performing testing as described in 3.2.

7.5 Subject term (key word) listing.

Amplifier
Emergency response
Law enforcement
Microphone
Public address system
Sound pressure level
Speaker
Wail
Yelp

MILITARY INTERESTS:	CIVIL AGENCY COORDINATING ACTIVITY:
Custodians: Air Force - 99 DLA - GS	GSA - FAS
Review Activity: DLA - IS	Preparing Activity: DLA - GS2
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